INCH-POUND

MIL-R-23417/3A
25 April 1991

SUPERSEDING
MIL-R-23417/3
11 September 1967

MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL, AC
TYPE 11R23W4A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Electrical Resolvers described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-R-23417.

TABLE I. Requirements

Requirements	Value	Unit	Tolerance
Excitation winding	Stator	---	---
Rated frequency	400	Hz	±1%
Input voltage:			
Maximum	26	volts (rms)	±1%
Test	10	volts (rms)	±1%
Minimum	0.5	volts (rms)	±1%
Primary current @ test voltage	5	milliamps	nominal
Primary power	8	milliwatts	nominal
D.C. resistance:			
Rotor	330	ohms	±15%
Stator	305	ohms	±15%
Compensator	350	ohms	±15%
Impedance:			
Zro	400 +j2325	ohms	R±15%+jx±10%
Zso	365 +j2200	ohms	R±15%+jx±10%
Zrs	610 +j250	ohms	reference
Zco	415 +j2150	ohms	R±15%+jk±10%
Zcs	650 +j10	ohms	reference
Resonant frequency:			
Rotor	55	kHz	minimum
Compensator	100	kHz	minimum
Low corner frequency	60	Hz	reference

AMSC - N/A

FSC 5990

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TABLE I. Requirements - Continued.

Requirements	Value	Unit	Tolerance
Transformation ratio:			
Output/input	0.980	----	±0.010
Rotor/compensator	0.990	----	±0.010
Transformation ratio variation (rotor/ compensator) due to voltage with test voltage as reference	0.5	percent	maximum
Equality of transformation ratio:			
Output/input	0.1	percent	maximum
Rotor/compensator	0.1	percent	maximum
Phase shift:			
Output/input	7.5	degrees	±1.5
Rotor/compensator	0	degrees	±0.15
Phase shift variation (Rotor/compensator) due to voltage with test voltage as reference	±4.0	minutes	maximum
Interaxis error	3.0	minutes	maximum
Functional error	0.1	percent	maximum
Null voltage:			
Fundamental	0.8	millivolts	maximum
Total	1.0	millivolts	maximum
Shift of resolver zero:			
With voltage	±1.5	minutes	maximum
With frequency	±0.5	minutes	maximum
Harmonic distortion	±0.1	percent	----
Temperature rise	5.0	degrees C	maximum
Mechanical breakaway torque:	0.07	oz-in	maximum
@ -55°C	0.15	oz-in	maximum
Radial play	0.0006	inches	maximum
End play	0.0010	inches	±0.0005

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TABLE II. Part identifying number variant characteristics.

Part Identifying No. 1/	A ±.010	B ±.010	G Maximum	Shaft Style	Terminal End	U ±.0000 -.0005
M23417/3-01A	.555	.500	1.800	Spline	Terminal Board	.1872
M23417/3-02A	.367	.312	1.550	Plain	Wire Leads	.1200
M23417/3-03A	.430	.375		Plain	Wire Leads	.1200
M23417/3-04A	.492	.437		Plain	Wire Leads	.1200
M23417/3-05A	.555	.500		Plain	Wire Leads	.1200
M23417/3-06A	.617	.562	1.550	Plain	Wire Leads	.1200

1/ Part identifying number M23417/3-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Part identifying number cross-reference. 1/

SUPERSEDED PART NUMBERS MIL-R-23417/3	NEW PART IDENTIFYING NUMBERS
M23417/3-01	M23417/3-01A
M23417/3-02	M23417/3-02A
M23417/3-03	M23417/3-03A
M23417/3-04	M23417/3-04A
M23417/3-05	M23417/3-05A
M23417/3-06	M23417/3-06A

1/ All line items shown in Table III refer to equivalent and interchangeable electrical resolvers of the same type designation modification. Part number changes do not affect form, fit or function of the electrical resolvers listed therein.

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NOTES:

1. Transformation ratio variation and phase shift variation (rotor/compensator) shall be tested as follows:

   a. Qualification: Variation shall remain within specification limits when subjected to individual or simultaneous variations of voltage (0.8 to 10 volts), frequency (380 to 420 Hz), and temperature (-55°C to +125°C). Test by applying input voltages of 0.8, 5.0, and 10 volts with frequency and temperature extremes simultaneously applied.

   b. Acceptance: Test by applying input voltages of 0.8, 5.0, and 10 volts at rated frequency and room ambient temperatures.

2. A voltage value of 0.5 to 26 volts shall be entered in the "volts" designation space on the identification plate.

Custodians:
Army - AR
Navy - AS
Air Force - 85

Review activities:
Army - MI
Navy - EC
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - AR

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